

EE5907 Pattern Recognition / EE5027 Statistical PR

Lecture 0 Intro and Admin Matters



National University of Singapore

Welcome to EE5907 / EE5027!

- ▶ Course will be conducted by Dr. XU Yuecong (1st part, till Recess) and (not for EE5027) Dr. Wang Si (2nd part, Wk 7 – 13).
- ▶ Topics on decision theory, linear/non-linear classifier and regressor, PCA, LDA, clustering, boosting, SVM, etc.
- ▶ In-person lectures:
 - ▶ Lecture notes will be available right before each class.
 - ▶ Recording will be available at the end of each class for revision.
 - ▶ Break for 15 mins in between, end in (less than) 2.5 hours (as long as content is completed). After class discussion welcomed!
- ▶ Q&A:
 - ▶ <https://docs.google.com/document/d/1qYrXugSIWwnOrG6H1P4yHLjrrlXrQ1q6uMR5S1-kbCk/edit?usp=sharing>

Meet Your Teaching Team

- ▶ Lecturers:
 - ▶ Dr. XU Yuecong (Part I, Week 1 – 6) yc.xu@nus.edu.sg
 - ▶ Dr. WANG Si (Part II, Week 7 – 13)
- ▶ Graduate Assistants (GA):
 - ▶ Nguyen Tuan Hiep e1439499@u.nus.edu

Background of Myself and Course

- Application of Models into Real-world Tasks and Scenarios
 - Focusing on Data-constrained and Computation-constrained environments
 - Focusing on specific application in Medical AI and Embodied AI
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- ▶ Pattern recognition and machine learning, not deep learning.
 - ▶ Understanding the concepts for application.
 - ▶ Some DL algorithms/techniques are based on PR/ML.

Learning Outcomes

- ▶ Comprehend principles and core techniques in pattern recognition such as Bayesian decision theory, linear and non-linear classifiers, principal component analysis, linear discriminant analysis, clustering and support vector machine.
- ▶ Apply the conceptual and mathematical tools of pattern recognition to analyze real-world data, obtaining data features and performing classification or regression.
- ▶ Assess the strengths and weaknesses of the different pattern recognition algorithms.

Assessment and Grading

- ▶ Continuous Assessment consists of 3 components.
 - ▶ CA1: Week 3 – Week 6, 27 Aug – 15 Sep: 15%.
 - ▶ CA1 To be announced on Week 3.
 - ▶ Quiz: 17 Sep (During lecture on Week 6): 5%.
 - ▶ Quiz format: MCQ, fully open book, max 60 minutes.
 - ▶ Quiz to be conducted via Canvas.
 - ▶ CA2: TBA by Dr. Wang Si: 20%.
- ▶ Final: 60%.
 - ▶ Closed book with one-page double-sided cheat sheet.

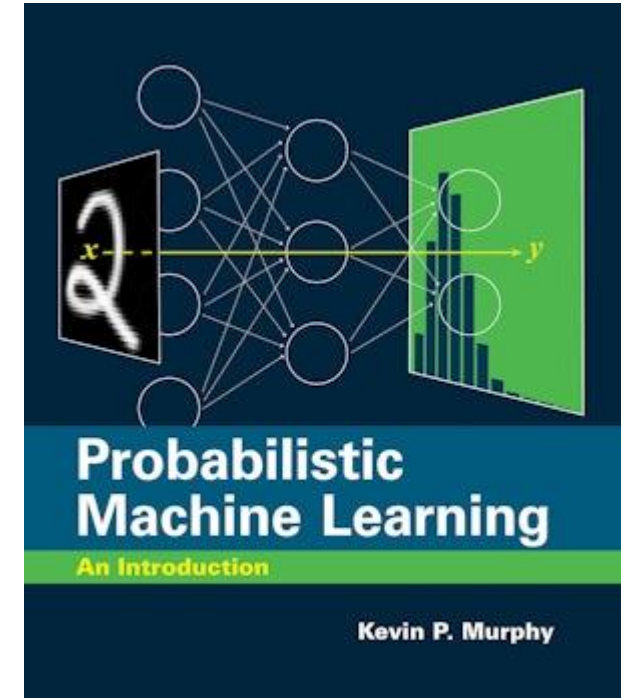
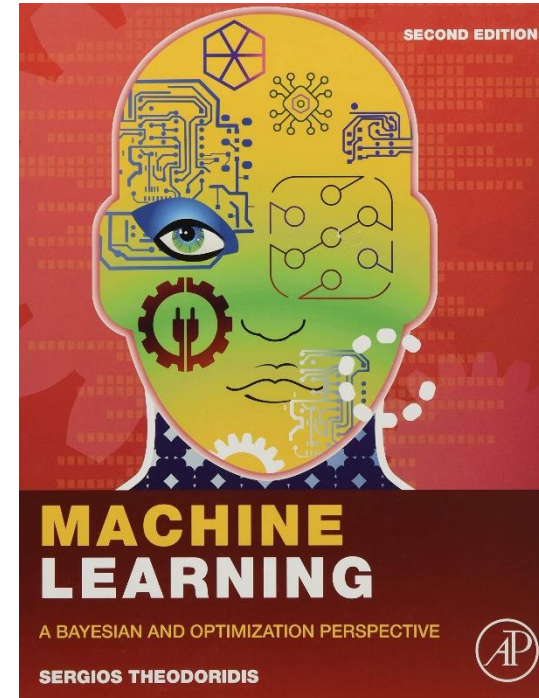
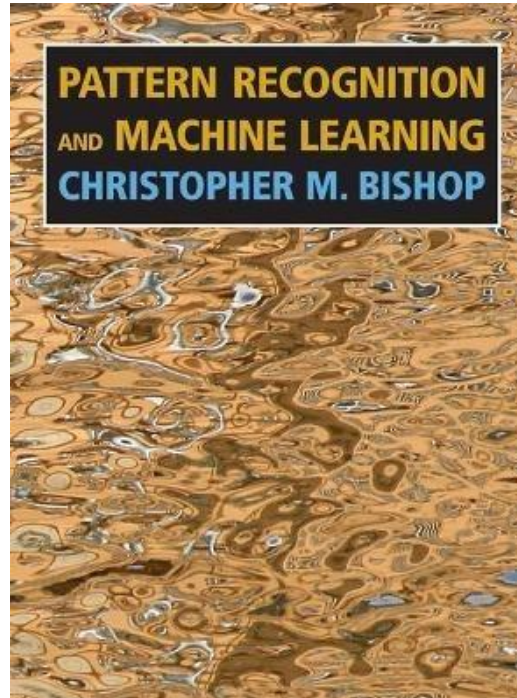
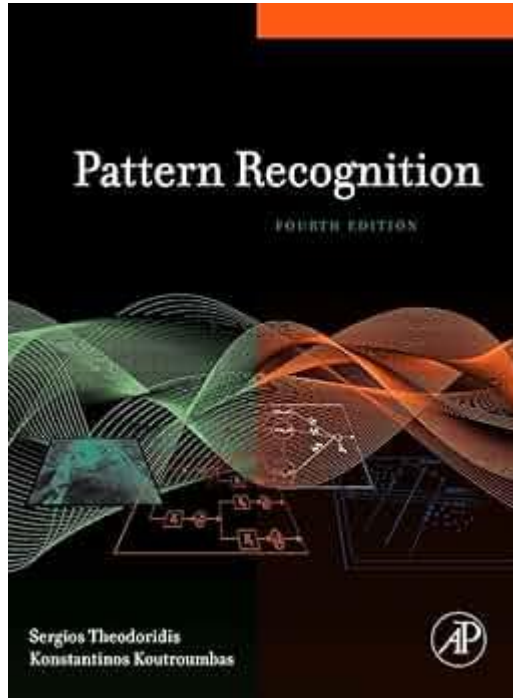
Assessment and Grading

- ▶ **Deadlines are strict.** Late submission will be deducted 10 points (out of 100) for every 24 hours. **ZERO** mark for late submission of 5 days (120 hours) or more.
- ▶ **NO plagiarism is allowed.** A **ZERO** mark will be given to anyone found plagiarizing and a formal report will be handed to the Department for further investigation.
- ▶ GPT-based/GPT-like tools (ChatGPT, Co-pilot, Gemini) allowed for refinement and coding (of course, amendments are required). They're **NOT allowed** for generating the whole project submission.

References

- ▶ Theodoridis, Sergios, and Konstantinos Koutroumbas. Pattern recognition. Elsevier, 2006.
- ▶ Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern recognition and machine learning. Vol. 4, no. 4. New York: springer, 2006.
- ▶ Theodoridis, Sergios. Machine Learning: A Bayesian and Optimization Perspective. Academic Press, 2020.
- ▶ Murphy, Kevin P. Probabilistic machine learning: an introduction. MIT press, 2022.

References



Code of Conduct

- ▶ Students are encouraged to actively participate.
 - ▶ Your grade depends on what you do, not what others do.
 - ▶ Don't share your solutions/codes. Submit only your own work.
- ▶ Academic Integrity:
 - ▶ Student's academic work & behavior are held to highest academic integrity standards. Cheating, plagiarism, unauthorized collaboration, and 'helping others commit these acts' are examples of academic misconduct, which can result in disciplinary action.
- ▶ **Any violation to academic honesty will be reported to the University's disciplinary panel.**

Tutorials

- ▶ Tutorials will be provided to help you understand the concepts better (and to better prepare for Finals 😊).
 - ▶ Not graded at all but will be relevant to the finals in the type of questions and the concepts that may be examined.
 - ▶ May not fully overlap but would be relevant.
 - ▶ Tutorial 1 will be provided on Week 3.
 - ▶ Tutorial 1 will be discussed on Week 4, during class.
- ▶ Tutorial 2 will be provided on Week 9.
- ▶ There will be NO discussion for Tutorial 2, but answers will be provided on Week 11.

Classroom Tools via Vercel

- ▶ We will be *experimenting* a new (top-up) classroom tool for learning, designed by myself, Dr. Chua Dingjuan (ECE) along with your seniors (*credits to them*).
- ▶ The tool is designed to improve understanding through in-class quizzes (not marked) which are generated and popped up in response to your progress in class.
- ▶ The slides are synced automatically, but you may turn to any pages that you want to read for a longer time anytime you want. You can click on “Sync” anytime you want.
- ▶ You may also raise comments / questions, click on “Too fast” or “Too slow” to advice the speed through the system.
- ▶ Due to technical issues, we are unable to save the questions and the answers for you, instead, please take screenshots when the questions and answers are available.
- ▶ Participation of this system is purely voluntary. Your identity will be strictly protected. Your participation or not of this system is NOT related to your scoring or learning process.